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Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25CST102 – STRUCTURAL DYNAMICS

(For CST)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Define the following: (i) Degree of Freedom, (ii) Damping Ratio, (iii) Resonance.	10	L2	1	1
b) Derive the equation of motion for a SDOF system using D'Alembert's principle.	10	L2	1	1
2. a) A steel cantilever beam is 1.5 m long, with a cross-section of 50 mm width and 10 mm depth. A mass of 50 kg is fixed at its free end. Calculate the natural frequency of the system. Neglect the mass of the beam. (E = 200 GPa).	10	L2	2	2
b) A spring-mass system has a natural frequency of 10 Hz. When 2 kg of mass is added, the frequency drops to 7.5 Hz. Find the original mass and the stiffness of the spring.	10	L3	2	3

Unit – II

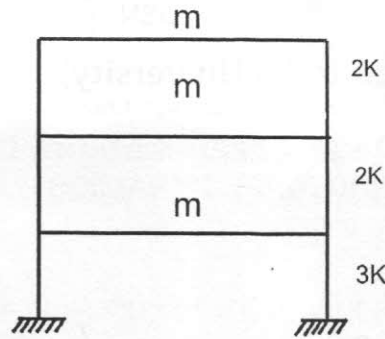
3. Derive equation of motion for damped free vibration of SDOF system and also derive equation of motion for critically damped, over damped and under damped system.	20	L6	2	3
4. a) How to measure Damping? List out different types of Dampers.	10	L6	2	3
b) Derive the response of an underdamped SDOF system to initial conditions (displacement and velocity). Sketch the typical response.	10	L6	2	3

Unit – III

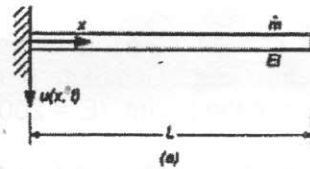
5. a) Derive the steady-state response of a damped SDOF system subjected to a harmonic force $F(t) = F_0 \sin(\omega t)$.	20	L5	3	3
6. a) Explain the terms 'Orthogonality of Modes' and 'Modal Participation Factor'.	10	L3	3	2
b) For a two-story shear building with identical floor masses (m) and identical story stiffnesses (k), determine the natural frequencies and mode shapes.	10	L5	3	3

Unit – IV

7. a) Explain Rayleigh's Method and derive equation for natural frequency.	10	L5	4	3
b) Determine by Rayleigh's Ritz Method, the fundamental frequency of the system shown in Figure Q 7(b). Given the mode shape as below $\{\phi\} = \begin{pmatrix} 1.00 \\ 0.80 \\ 0.40 \end{pmatrix}$	10	L5	4	3



8. Find the natural frequency of vibration of the cantilever beam shown in figure using shape function $\psi_0(x) = (x/L)^2$ obtain more accurate estimate of the frequency by Improved Rayleigh's Method.



Unit – V

(a)		20	L5	4	3	
Unit – V						
9.	a)	What are the principles of Vibration Isolation? Differentiate between active and passive isolation.	10	L2	4	3
	b)	What are flexural vibrations? write natural vibration and mode shapes for simple supported beams.	10	L2	4	3
10.	a)	Write a short note on earthquake measuring equipment's.	10	L4	5	4
	b)	Discuss any one laboratory experiments in dynamics in detail.	10	L4	5	4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
